1. Clinical Question & Core Hypothesis

Clinical Question

The graph links PTEN status to prognosis and separately links LAG3 to prognosis, suggesting a tension where immune infiltration (TILs) does not always yield better outcomes; mechanistically, PTEN loss activates PI3K/AKT/mTOR signaling which could reprogram the tumor microenvironment to favor T-cell dysfunction (LAG3-mediated exhaustion). The key question becomes: why do some TIL-rich tumors still have poor survival, and could PTEN-driven signaling explain elevated LAG3-related immune suppression that nullifies TIL benefit?

Working Hypothesis

In breast cancer, PTEN loss leads to heightened PI3K/AKT/mTOR pathway activity that reshapes the tumor microenvironment toward immunosuppression. When intratumoral TIL density is high, concurrent high LAG3 expression indicates T-cell exhaustion, which dampens the beneficial prognostic effect of TILs. Therefore, PTEN-deficient, TIL-high, LAG3-high tumors will have worse overall survival than PTEN-intact, TIL-high, LAG3-low tumors. The hypothesis remains testable across TCGA-BRCA data and can be interrogated for potential therapeutic synergy with mTOR pathway inhibition and/or LAG3 blockade, with TP53 and mTORC1 status acting as secondary modifiers that influence the strength of the PTEN–LAG3–TIL interaction.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PTEN, LAG3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTEN: PTEN status to prognosis and separately links LAG3 to prognosis, suggesting a tension where immune infiltration (TILs) does not always yield better outcomes; mechanistically, PTEN loss activates PI3K/AKT/mTOR signaling which could reprogram the tumor microenvironment to favor T-cell dysfunction (LAG3-mediated exhaustion).

LAG3: LAG3 to prognosis, suggesting a tension where immune infiltration (TILs) does not always yield better outcomes; mechanistically, PTEN loss activates PI3K/AKT/mTOR signaling which could reprogram the tumor microenvironment to favor T-cell dysfunction (LAG3-mediated exhaustion).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph

Secondary	mTORC1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: directly link tumor-intrinsic signaling to an immune checkpoint axis, with TIL density serving as a functional readout of immune infiltration. Overall survival (OS) is a robust, clinically meaningful endpoint available in TCGA-BRCA, making it suitable to capture the net effect of the PTEN–LAG3–TIL interaction on outcome.

Prediction 2: . Co-culture with activated T cells to measure T cell activation/exhaustion markers (LAG3, PD-1, TIM-

Prediction 3: , cytokine output, and cytotoxicity; test effects of rapamycin or TORC1 inhibitors and LAG3 blockade. - In vivo validation: Use PTEN-knockdown breast cancer syngeneic models to evaluate tumor growth, TIL density, LAG3 expression, and response to combined mTOR inhibition and LAG3 blockade vs single modalities, with survival as an endpoint. - Outcome relevance: Demonstrate that the PTEN–LAG3–TIL axis identifies a prognostically poor subset and provide a preclinical rationale for combined metabolic and immune checkpoint therapies in PTEN-deficient breast cancer.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- mTORC1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.